

Seminar On Algebraic Geometry
Lecture Notes 2 (Incomplete version)

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In this lecture, we will discuss sheaves of abelian groups (moreover, sheaves of modules) by viewing Sh as an abelian category.

1 Abelian Category Of Sheaves

Let X be a topological space.

Definition 1.1. Let $\mathscr{F}'$ be a subsheaf of a sheaf $\mathscr{F}$, i.e. there is an injective morphism $i : \mathscr{F}' \to \mathscr{F}$ (we can view $\mathscr{F}'(U)$ as a subsheaf of $\mathscr{F}(U)$ for each open subset U). We define the quotient sheaf $\mathscr{F}/\mathscr{F}'$ to be the sheafification of the presheaf $U \mapsto \mathscr{F}(U)/\mathscr{F}'(U)$. For any $P \in X$, we have $(\mathscr{F}/\mathscr{F}')_P = \mathscr{F}_P/\mathscr{F}'_P$.

Corollary 1.2. Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a morphism of sheaves. Then $\operatorname{im}\varphi \cong \mathscr{F}/\ker\varphi$, and $\operatorname{coker}\varphi \cong \mathscr{G}/\operatorname{im}\varphi$.

Proposition 1.3 (Direct sum)**.**

Proposition 1.4 (Direct limit)**.**

Proposition 1.5 (Inverse limit)**.**

2 Operations on Sheaves

2.1 Global Section Functor

Recall that we sometimes write $\Gamma(U, \mathscr{F}) := \mathscr{F}(U)$, for any open subset $U \subseteq X$ and any sheaf $\mathscr{F}$ on X.

Proposition 2.1. [Ha, Ex 1.8] Suppose $U \subseteq X$ is an open subset, and $0 \to \mathscr{F} \to \mathscr{G} \to \mathscr{H}$ is an exact sequence of sheaves of abelian groups. Then

$$0 \to \Gamma(U, \mathscr{F}) \to \Gamma(U, \mathscr{G}) \to \Gamma(U, \mathscr{H})$$

is exact. I.e., $\Gamma(U, -)$ is a left exact functor $\mathrm{Sh}(X) \to \mathrm{Ab}$.

Proof. We can easily check this on the level of stalks, since for any morphism $\varphi : \mathscr{F} \to \mathscr{G}$ of sheaves, and for any $P \in U$, we have $\ker\varphi_P = (\ker\varphi)_P$ and $\operatorname{im}\varphi_P = (\operatorname{im}\varphi)_P$. □

However, $\Gamma(U, -)$ is not exact in general, especially in algebraic geometry: a surjective morphism of sheaves need not induce a surjection on sections. The following two examples illustrate that differential topology is "soft," whereas the Zariski topology is "hard."

Lemma 2.2. Let M be a smooth (paracompact) manifold and let $\{U_i\}_{i\in I}$ be an open cover of M. Then there exists a locally finite partition of unity subordinate to $\{U_i\}$: a family $\{\rho_i\}_{i\in I}$ of smooth functions $\rho_i \in C^\infty(M)$ such that $0 \le \rho_i \le 1$, $\operatorname{supp}\rho_i \subseteq U_i$ is compact, the family $\{\operatorname{supp}\rho_i\}_{i\in I}$ is locally finite, and $\sum_{i\in I}\rho_i = 1$ on M (the sum is locally finite, hence well defined).

Example 2.3. Let M be a smooth manifold, $\mathscr{C}^\infty_M$ the sheaf of smooth functions on M, and $\mathscr{C}^\infty_{M,c}$ the presheaf of compactly supported smooth functions. The inclusion $\mathscr{C}^\infty_{M,c} \hookrightarrow \mathscr{C}^\infty_M$ induces an isomorphism

$$(\mathscr{C}^\infty_{M,c})^{\mathrm{sh}} \xrightarrow{\sim} \mathscr{C}^\infty_M$$

on sheafification. Indeed, for each $f \in C^\infty(M)$ and each point, choose a locally finite open cover $\{U_i\}_{i\in I}$ of M by relatively compact open subsets and a partition of unity $\{\rho_i\}_{i\in I}$ subordinate to it. Then each $f_i := \rho_i f$ is compactly supported and only finitely many f_i are nonzero near each point, so $f = \sum_i f_i$ with $\sum_i f_i \in \Gamma(M, \mathscr{C}^\infty_{M,c})$. Thus every smooth function is locally a sum of compactly supported smooth functions, and the sheafification of $\mathscr{C}^\infty_{M,c}$ is $\mathscr{C}^\infty_M$.

Proposition 2.4. Let M be a smooth manifold and $\mathscr{O}_M := \mathscr{C}^\infty_M$ the sheaf of smooth functions. Then the global section functor

$$\Gamma(M, -) : \mathscr{O}_M\text{-}\mathsf{Mod} \longrightarrow \mathsf{Ab}$$

is exact.

Proof. Since $\Gamma(M, -)$ is left exact, it suffices to show that every surjective morphism $\varphi : \mathscr{F} \to \mathscr{G}$ of $\mathscr{O}_M$-modules induces a surjection on global sections. Let $g \in \Gamma(M, \mathscr{G})$. As φ is an epimorphism of sheaves, there exist an open cover $\{U_i\}_{i\in I}$ of M and local lifts $f_i \in \mathscr{F}(U_i)$ with $\varphi(f_i) = g|_{U_i}$. Choose a partition of unity $\{\rho_i\}_{i\in I}$ subordinate to $\{U_i\}_{i\in I}$. Since $\mathscr{F}$ is an $\mathscr{O}_M$-module, the section $\rho_i f_i \in \mathscr{F}(U_i)$ vanishes near the boundary of U_i, hence extends by zero to a global section $\widetilde{\rho_i f_i} \in \Gamma(M, \mathscr{F})$. By local

finiteness the sum $f := \sum_{i\in I} \widetilde{\rho_i f_i}$ is a well-defined global section of $\mathscr{F}$, and

$$\varphi(f) = \sum_{i\in I} \rho_i\, \varphi(f_i) = \sum_{i\in I} \rho_i\, g|_{U_i} = g.$$

Hence φ is surjective on global sections, and $\Gamma(M, -)$ is exact. □

Example 2.5. [Va, Example 2.4.10] Let $X = \mathbb{C}$ with the classical topology, and define $\mathscr{O}_X$ to be the sheaf of holomorphic functions, and $\mathscr{O}_X^*$ to be the sheaf of invertible holomorphic functions. Then we have an exact sequence

$$0 \longrightarrow \underline{\mathbb{Z}} \xrightarrow{\times 2\pi i} \mathscr{O}_X \xrightarrow{\exp} \mathscr{O}_X^* \longrightarrow 1,$$

where $\underline{\mathbb{Z}}$ is the constant sheaf associated to $\mathbb{Z}$.

Indeed, we can check this on stalks. Without loss of generality, consider $P = 0/$ Then $\underline{\mathbb{Z}}_0 = \mathbb{Z}$, and $\mathscr{O}_{X,0}$ is the ring of convergent Taylor series at origin, and $\mathscr{O}_{X,0}^*$ is the ring of functions in $\mathscr{O}_{X,0}$ which doesn't vanish at 0. The image of $\times 2\pi i$ on stalks is $2\pi i\mathbb{Z}$ as constant functions, and the kernel of exp is just $2\pi i\mathbb{Z}$ by Euler's equation. To show that the expotential map is surjective on stalks, we only need to notice that for any holomorphic function $f \in \mathscr{O}_{X,0}^*$, there is an open neighborhood U of the origin such that f has no roots in U. Thus, $\ln f$ is a holomorphic function in U, which gives the preimage of f in $\mathscr{O}_{X,0}$.

Let $U = \mathbb{C} \setminus \{0\}$ be the punctured complex plane. Suppose that there is some $f \in \mathscr{O}_X(U)$, such that $\exp f = \mathrm{id}_U \in \mathscr{O}_X^*(U)$, i.e., $\exp(f(z)) = z$ on U. Then $f'(z)z = f'(z)e^{f(z)} = 1$, for $z \in U$, and we have $f'(z) = 1/z$. Now we calculate the integral at $\gamma : |z| = 1$:

$$\oint_\gamma f'(z)\,\mathrm{d}z = \oint_\gamma \frac{1}{z}\,\mathrm{d}z = 2\pi i \neq 0.$$

However, by Cauchy's Integral Theorem, since γ is a closed curve containing in U, a holomorphic open domain of $f'(z)$, the integral must be zero. Therefore, such f does not exist, and $\exp(U)$ is not surjective.

2.2 Hom Sheaf

Proposition 2.6. Let $\mathscr{F}, \mathscr{G}$ be sheaves of abelian groups on X. For any open set $U \subseteq X$, the set $\mathscr{H}om\,(\mathscr{F}\,|_U, \mathscr{G}\,|_U)$ is an abelian group, and $\mathscr{H}om(\mathscr{F}, \mathscr{G})$ is a sheaf of abelian groups on X, by $U \mapsto \mathrm{Hom}\,(\mathscr{F}\,|_U, \mathscr{G}\,|_U)$. It is called the *sheaf of local morphisms* of $\mathscr{F}$ into $\mathscr{G}$, "sheaf hom" for short, and is denoted

$\mathrm{Hom}\,(\mathscr{F},\mathscr{G})$.

Proof. For each open $U \subseteq X$, the set $\mathrm{Hom}(\mathscr{F}\mid_U, \mathscr{G}\mid_U)$ becomes an abelian group under pointwise addition: for morphisms φ, ψ and any open $V \subseteq U$, $s \in \mathscr{F}(V)$, put

$$(\varphi + \psi)(V)(s) = \varphi(V)(s) + \psi(V)(s),$$

and define $(-\varphi)(V)(s) = -\varphi(V)(s)$, the zero morphism sending every section to 0. The group axioms are inherited from the abelian groups $\mathscr{G}(V)$, so $U \mapsto \mathrm{Hom}(\mathscr{F}\mid_U, \mathscr{G}\mid_U)$ is a presheaf of abelian groups.

It remains to check the sheaf axioms. Let $U \subseteq X$ be open and $\{U_i\}_{i\in I}$ an open cover of U. *Identity:* if $\varphi \in \mathrm{Hom}(\mathscr{F}\mid_U, \mathscr{G}\mid_U)$ satisfies $\varphi|_{U_i} = 0$ for all i, then for every open $V \subseteq U$ and $s \in \mathscr{F}(V)$ the sections $\varphi(V)(s)|_{V\cap U_i} = 0$ for all i; since $\mathscr{G}$ is a sheaf and $\{V \cap U_i\}_{i\in I}$ covers V, we get $\varphi(V)(s) = 0$, hence $\varphi = 0$. *Gluing:* suppose $\varphi_i \in \mathrm{Hom}(\mathscr{F}\mid_{U_i}, \mathscr{G}\mid_{U_i})$ satisfy $\varphi_i|_{U_i\cap U_j} = \varphi_j|_{U_i\cap U_j}$ for all i, j. For open $V \subseteq U$ and $s \in \mathscr{F}(V)$, the sections $\varphi_i(V \cap U_i)(s|_{V\cap U_i}) \in \mathscr{G}(V \cap U_i)$ agree on the overlaps $V \cap U_i \cap U_j$; by the sheaf property of $\mathscr{G}$ they glue to a unique section $\varphi(V)(s) \in \mathscr{G}(V)$. The assignments $V \mapsto \varphi(V)$ define a morphism of sheaves $\mathscr{F}\mid_U \to \mathscr{G}\mid_U$, and $\varphi|_{U_i} = \varphi_i$ by construction. Hence the presheaf is a sheaf of abelian groups. □

Proposition 2.7. Suppose $\mathscr{F}$ is a sheaf of abelian groups on a topological space X. Then $\mathscr{H}om\,(\mathscr{F},\cdot)$ is a left-exact functor $\mathrm{Sh}(X) \to \mathrm{Sh}(X)$, a nd $\mathscr{H}om\,(\cdot,\mathscr{F})$ is a left-exact contravariant functor $\mathrm{Sh}(X) \to \mathrm{Sh}(X)$, as we define above.

Proof. Since the Hom sheaf is defined on sections by $\mathscr{H}om(\mathscr{F},\mathscr{G})(U) = \mathrm{Hom}(\mathscr{F}\mid_U, \mathscr{G}\mid_U)$, and exactness of a sequence of sheaves is checked on every open set (restriction to an open subset preserves exactness), it suffices to prove the corresponding exactness of Hom in $\mathrm{Sh}(U)$ for each open U.

Let $0 \to \mathscr{G}' \xrightarrow{\alpha} \mathscr{G} \xrightarrow{\beta} \mathscr{G}''$ be exact. We show that

$$0 \to \mathrm{Hom}(\mathscr{F},\mathscr{G}') \xrightarrow{\alpha_*} \mathrm{Hom}(\mathscr{F},\mathscr{G}) \xrightarrow{\beta_*} \mathrm{Hom}(\mathscr{F},\mathscr{G}'')$$

is exact. If $\alpha \circ \varphi = 0$, then $\varphi = 0$ since α is injective; hence α_* is injective. As $\beta \circ \alpha = 0$, we have $\beta_* \circ \alpha_* = 0$. Conversely, if $\beta \circ \varphi = 0$, then φ factors through $\ker\beta = \mathrm{im}\,\alpha \cong \mathscr{G}'$ (using that α is injective), so $\varphi = \alpha \circ \psi$ for a unique $\psi \in \mathrm{Hom}(\mathscr{F},\mathscr{G}')$. Thus $\ker\beta_* = \mathrm{im}\,\alpha_*$, proving left-exactness of $\mathscr{H}om(\mathscr{F},\cdot)$. □

3 Pushforward & Inverse Image

Definition 3.1. Let $f : X \to Y$ be a continuous map of topological spaces.

- For any sheaf $\mathscr{F}$ on X, we define the *direct image (pushforward)* sheaf $f_*\mathscr{F}$ on Y by
$$f_*\mathscr{F}(V) := \mathscr{F}(f^{-1}V),$$
for any open set $V \subseteq Y$.
- For any sheaf $\mathscr{G}$ on Y, we define a presheaf $f^{-1}\mathscr{G}$ on X by
$$U \mapsto \varinjlim_{V \supseteq f(U)} \mathscr{G}(V),$$
for any open set $U \subseteq X$. And we define *inverse image* sheaf $f^{-1}\mathscr{G}$ to be the sheafification of the presheaf given above.

Then $f_* : \mathrm{Sh}(X) \to \mathrm{Sh}(Y)$ and $f^{-1} : \mathrm{Sh}(Y) \to \mathrm{Sh}(X)$ are two functors between sheaves on different topological spaces.

Proposition 3.2. Let $f : X \to Y$ be a continuous map of topological space. Then $f^{-1} : \mathrm{Sh}(Y) \to \mathrm{Sh}(X)$ is left-adjoint to $f_* : \mathrm{Sh}(X) \to \mathrm{Sh}(Y)$. Precisely, there is a natural isomorphism
$$\tau_{\mathscr{F}\mathscr{G}} : \mathrm{Hom}_X\left(f^{-1}\mathscr{G}, \mathscr{F}\right) \to \mathrm{Hom}_Y\left(\mathscr{G}, f_*\mathscr{F}\right)$$
for any sheaf $\mathscr{F} \in \mathrm{Sh}(X)$ and $\mathscr{G} \in \mathrm{Sh}(Y)$.

Definition 3.3. If Z is a subset of X, regarded as a topological subspace with the induced topology , if $i : Z \to X$ is the inclusion map, and if $\mathscr{F}$ is a sheaf on X, then we call $i^{-1}\mathscr{F}$ the *restriction* of $\mathscr{F}$ to Z, and we often denote it by $\mathscr{F}\mid_Z$. If U is an open subset of X, then $\mathscr{F}\mid_U$ is naturally defined by $\mathscr{F}\mid_U(V) = \mathscr{F}(V)$ for any open subset $V \subseteq U$.

Lemma 3.4. Let Z be a subset of X, $\mathscr{F}$ be a sheaf on X. Then $(\mathscr{F}\mid_Z)_P = \mathscr{F}_P$ for any point $P \in Z$.

Example 3.5 (Skyscraper)**.**

4 Sheaves Of Modules: An Introduction

5 Base Sheaves

6 Gluing Sheaves

References

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[Va] Ravi Vakil. *The Rising Sea: Foundations of Algebraic Geometry*. Princeton University Press, 2025. ISBN: 978-0691268682.